Analyzing and Mitigating Dataset Artifacts via Contrast Sets and Dropout Based Representation Cartography

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Abstract

This paper investigates dataset artifacts in the Stanford Natural Language Inference (SNLI) dataset using contrast set analysis and dropouts based Bayesian approximation dataset cartography. I train the ELECTRA-small model on the SNLI dataset and evaluate its robustness on a synthetically generated contrast validation set created by introducing controlled perturbations through lexical antonym, synonym, and negation edits using the NLTK library. Evaluation reveals a substantial degradation in model accuracy across all perturbation types. To analyze model behavior, I employ a modified dataset cartography approach inspired by Swayamdipta et al. (2020), applying Monte Carlo Dropout during evaluation to estimate model uncertainty and variability, following Gal and Ghahramani (2016) approach, enabling efficient uncertainty estimation on small validation data with pre-trained models rather than during training as described in the original method by *Swayamdipta et al paper*. Visualization shows greater dispersion and ambiguity in edited examples, highlighting sensitivity to linguistic variation. Fine-tuning on a mixture of contrast examples and original data improves model robustness and generalization to unseen contrast data, offering insights into mitigating dataset artifacts. Additionally, this approach shows promising performance, although there is still room for improvement on the unseen edited test split. To better understand this discrepancy, I further apply a CoLA-based grammatical acceptability classifier (Warstadt et al., 2019; Morris et al., 2020) to assess the linguistic quality of the generated contrast hypotheses. Finally, the findings motivate future work exploring instruction-tuned LLMs to generate more linguistically coherent and semantically controlled contrast sets.

1 Introduction

This paper investigates dataset artifacts in the Stanford Natural Language Inference (SNLI) dataset using contrast set analysis and dataset cartography. I train the ELECTRA-small model on the SNLI dataset and assess its performance by evaluating it on a synthetically generated contrast validation set. This set contains controlled perturbations introduced through lexical antonym, synonym, and negation edits, all created using the NLTK library. A notable drop in model accuracy is observed across all types of perturbations, with synonym edits presenting especially tough challenges. To comprehensively analyze model behavior, I adopt a modified dataset cartography approach, inspired by Swayamdipta et al. (2020). This modified approach incorporates Monte Carlo Dropout during evaluation to approximate model uncertainty and variability, as suggested by Gal and Ghahramani (2016). This enables efficient uncertainty estimation on small validation datasets for pre-trained models, unlike the original method that employed this technique during training. Visualization shows greater dispersion and ambiguity in edited examples, highlighting sensitivity to linguistic variation. Fine-tuning the model on a combination of contrast examples and a portion of the original data enhances its robustness and generalization to unseen contrast data. This approach provides precious knowledge into mitigating dataset artifacts in natural language inference.

Natural Language Inference (NLI) is a core benchmark for evaluating a model's ability to reason about entailment and contradiction between textual pairs. The pre-trained language models such as BERT and ELECTRA achieve strong performance on benchmark datasets like the Stanford Natural Language Inference (SNLI) corpus. However, recent work (e.g., (Gardner et al., 2020)) has shown that such datasets often contain spurious artifacts and annotation biases, leading models to exploit superficial lexical patterns rather than learning genuine semantic reasoning. As a result, models that perform well on the original valida-

tion or test splits may struggle when evaluated on minimally perturbed examples, revealing limited robustness and generalization.

Contrast set evaluation (Gardner et al., 2020) is one of the diagnostic framework to measure a model's stability under small, semantically meaningful edits to the input text. By constructing pairs of original and modified examples that differ minimally along with flipping to its correct label, contrast sets expose brittle model behavior and hidden biases in both data and model predictions, confirming the model's reliance on shallow lexical cues. However, constructing and analyzing such contrast data at scale remains a challenge. My work aims to systematically and at scale generate this contrast set. Additionally, I explore how visualization and representation techniques, such as dataset cartography Swayamdipta et al. (2020), can complement this analysis..

To better understand these behavioral shifts, I adopt and extend the dataset cartography framework. Unlike prior work, which computes training dynamics during model training, I apply Monte Carlo Dropout during evaluation as a Bayesian approximation Gal and Ghahramani (2016) to estimate prediction confidence and variability on SNLI validation data split. This modified approach enables efficient cartography analysis of model uncertainty and variability for pre-trained models without retraining on massive datasets.

Finally, I explore whether fine-tuning on a mixture of contrast examples and a subset of the original training data can recover performance. The results demonstrate improved robustness on both contrast validation and unseen contrast SNLI test sets. The findings provide insights into which perturbation types remain most challenging and suggest a scalable method for incorporating linguistically diverse contrast data into NLI training pipelines. To better understand the uneven robustness improvements and evaluate whether synthetic edits were grammatically correct, I additionally assess the grammatical acceptability of all generated hypotheses using a CoLA-based classifier (Warstadt et al., 2019; Morris et al., 2020), motivating future work that explores instruction-tuned LLMs for generating more syntactically reliable and semantically controlled contrast sets.

2 Related Work

This project builds upon three key strands of prior work.

First, (Gardner et al., 2020) introduced contrast sets as a diagnostic framework for evaluating model robustness under small but semantically meaningful input perturbations. Their work demonstrated that models often fail on minimally edited examples, revealing reliance on dataset artifacts rather than true semantic understanding.

Second, Swayamdipta et al. (2020) proposed dataset cartography, a visualization framework for analyzing training dynamics through confidence and variability, identifying easy, ambiguous, and hard examples within a dataset. This approach offers an interpretable lens into how models learn and generalize by analyzing confidence and prediction variability across training.

Finally, (Gardner et al., 2020) showed that Monte Carlo Dropout can serve as a Bayesian approximation for estimating predictive uncertainty. Inspired by this, my project adapts the cartography framework to apply Monte Carlo Dropout during evaluation rather than training, to efficiently approximate uncertainty for pre-trained models with small dataset and visualize model behavior on contrast sets.

3 Method / Approach

This section outlines the methodology used to investigate dataset artifacts in the SNLI dataset, integrating six main components: (1) baseline model training, (2) contrast set construction, (3) accuracy comparison across edit types, (4) modified dataset cartography - Dropout based Bayesian inference approximation , (5) fine-tuning for robustness recovery, and (6) Synthetic contrast set analysis. To maintain clarity, I include only the key figures and tables in the main paper. Additional visualizations—such as test split performance plots, provided in the Appendix.

3.1 Baseline model training

To establish a robust baseline, I adopted the provided project starter code, which implements a standardized fine-tuning pipeline for natural language inference tasks. I used the *ELECTRA-Small*(Clark et al., 2020) model and trained on the Stanford Natural Language Inference (SNLI) dataset (Bowman et al., 2015).

3.2 Contrast Set Construction

To assess model robustness to linguistic variation, I generated a *contrast validation set* by minimally editing examples from the SNLI validation split. Following the idea of contrast sets proposed by Gardner et al. (2020), contrast examples were created through controlled perturbations while preserving grammatical and contextual validity. Edits were implemented using the NLTK library and included: (1) lexical antonym substitutions, which invert polarity or meaning; (2) lexical synonym replacements, maintaining semantic similarity; and (3) negation insertions, reversing entailment or contradiction relationships.

Unlike Gardner et al. (2020), who used manually curated contrast sets, this work automatically synthesizes contrast data through rule-based perturbations and then sanity proof read handful of them of each category. In total, the validation examples of 48% (4,706) were minimally modified. The composition of the resulting dataset is summarized in Table 1.

Edit Type	Description	Portion (%)
Antonym	Polarity inversion	11
Synonym	Meaning-preserving	18
Negation	Inserted negation	19
No Edit	Original, unmodified	52

Table 1: Composition of the synthetic SNLI contrast validation set minimal edits.

Each edited hypothesis was paired with its original counterpart to form a semantically related contrast set. This design enables analysis of prediction consistency under minimal but meaning-altering perturbations.

3.3 Accuracy Comparison Across Edit Types

Before introducing the uncertainty-based cartography analysis, I first quantify the robustness gap across different linguistic perturbations. The ELECTRA-small model, trained on the original SNLI dataset, achieved 89% accuracy on the unedited validation set. However, Its performance on the contrast validation set dropped substantially across the three perturbation categories.

The results in Table 2 reveal that antonym substitutions lead to the largest degradation in performance, followed by negation and synonym edits. This step highlights which linguistic phenomena expose model vulnerabilities, motivating deeper analysis using dataset cartography.

Edit Type	Original Accuracy (%)	Accuracy (%)
Antonym	89	40
Synonym	89	75
Negation	89	66

Table 2: Accuracy comparison between original validation examples and their contrast counterparts for each edit type.

3.4 Modified dataset cartography - Dropout based Bayesian inference approximation

To further analyze model behavior, I employ a modified dataset cartography approach inspired by Swayamdipta et al. (2020). While the original method computes confidence, variability, and correctness statistics over multiple training epochs, I adapt it to pre-trained models with small validation set during inference by estimating uncertainty using Monte Carlo Dropout Bayesian approximation (Gal and Ghahramani, 2016).

Under this approximation, dropout remains active during inference, and multiple stochastic forward passes are performed for each input. Let $f(\mathbf{x}; \mathbf{W})$ denote the model output for input $\mathbf{x}$ with dropout-induced weight configuration $\mathbf{W}$. The predictive distribution is approximated as:

$$p(y \mid \mathbf{x}, \mathcal{D}) \approx \frac{1}{T} \sum_{t=1}^{T} p(y \mid \mathbf{x}, \mathbf{W}_t), \qquad (1)$$

where T is the number of stochastic forward passes, and $\mathbf{W}_t \sim q(\mathbf{W})$ represents a random dropout mask.

The predictive mean serves as model confidence, while the predictive variance estimates uncertainty:

$$\mathrm{Var}[p(y \mid \mathbf{x})] = \frac{1}{T} \sum_{t=1}^{T} (p_t(y \mid \mathbf{x}) - \bar{p}(y \mid \mathbf{x}))^2. \qquad (2)$$

These uncertainty estimates are used to construct a cartography map plotting confidence and uncertainty. Unlike Swayamdipta et al. (2020), this adaptation avoids repeated full-epoch retraining, offering an efficient uncertainty-based visualization suitable for pre-trained transformer models such as ELECTRA-small with validation set.

3.5 Fine-Tuning for Robustness

To recover performance degraded by linguistic perturbations, I fine-tune the ELECTRA-small

Dataset Component	Examples	Description
Contrast Edits	∼5k	Edits (Negation, Antonym, Synonym)
Training Subset	∼87k	15% of original SNLI training data
Total	∼92k	Combined dataset

Table 3: Composition of the training dataset used for fine-tuning ELECTRA-small to mitigate performance drop on contrast sets.

model on a combined dataset. The training set for fine-tuning is prepared as follows:

This hybrid setup ensures balanced exposure to original and perturbed samples. The resulting fine-tuned model demonstrates improved robustness on both the contrast validation set and unseen test splits, validating that incorporating contrastive examples during fine-tuning helps mitigate spurious correlations of dataset artifacts.

3.6 Synthetic contrast set analysis

To analyze the linguistic quality of the synthetically generated contrast hypotheses, I applied a grammatical acceptability classifier based on the CoLA dataset (Warstadt et al., 2019). Specifically, I used the *textattack/roberta-base-CoLA* model (Morris et al., 2020) to score each edited hypothesis and classify it as grammatically acceptable or not. This automated check provides a scalable alternative to manual validation, enabling systematic quality assessment across all edit types.

4 Implementation / Experimental Setup

All commands used for each step of the implementation are provided in Appendix A.

4.1 Baseline Training

I trained the *ELECTRA-small* model (Clark et al., 2020) on the *SNLI* dataset (Bowman et al., 2015) using the Hugging Face Transformers library provided in project starter code. The model was initially trained on the original SNLI training split, achieving an accuracy of 89% on the validation set and 88% on test split. The training procedure follows the conventional sequence-pair classification setup, where each premise–hypothesis pair is encoded jointly and passed through the discriminator head to predict entailment, contradiction, or neutral.

4.2 Contrast set construction

A contrast validation set of ∼10k examples were synthesized by minimally editing hypotheses using the NLTK library. The contrast dataset composition is given in Table 1. The contrast test set was also created with a similar composition.

4.3 Data Analysis and Evaluation

Evaluation was performed on both the original and contrast validation and test sets to assess model robustness against minimal edits. The accuracies comparison is done for each edit type to understand the accuracy drop. Furthermore, Monte Carlo Dropout was employed as a Bayesian approximation (Gal and Ghahramani, 2016) during the evaluation phase to estimate model uncertainty, following the data cartography methodology proposed by Swayamdipta et al. (2020).

4.4 Performance Enhancement

The model is fine-tuned for about one epoch on mixed of contrast validation set and original SNLI training examples. The mixed dataset strategy is designed to mitigate overfitting on the edited contrast data. The fine-tuning set combined all edited examples ˜5k with an additional 15% subset of the original SNLI training data (˜87k examples), resulting in a total of ˜92k samples used for fine-tuning.

4.5 Analysis of synthetic contrast set

To estimate the linguistic quality of the generated contrast hypotheses, I applied an grammatical acceptability classifier based on the CoLA dataset (Warstadt et al., 2019). Specifically, I used the *textattack/roberta-base-CoLA* model (Morris et al., 2020) to score each edited hypothesis and label it as grammatically correct if its predicted acceptability probability exceeded the threshold. This procedure allowed for a scalable method, instead of a manual sanity check, to assess the syntactic correctness of all edit types within the contrast set.

5 Results

I first evaluate the *ELECTRA-small* model trained on the SNLI training dataset to establish a baseline. On the original validation and test split, the model achieves an high accuracy. However, when evaluated on the synthetically generated *contrast*

validation and test set, performance drops significantly are shown in Table 4, highlighting the model's sensitivity to minimal lexical and logical perturbations.

SNLI Spilt	Original Accuracy (%)	Contract Accuracy (%)
Validation set	89	77
Test set	88	76

Table 4: Overall Accuracy comparison between the original and contrast sets.

Table 5 summarizes the accuracy breakdown across different perturbation types in validation and test sets. Among the three edit categories, *lexical antonym* substitutions caused the largest degradation, while *synonym* and *negation* edits showed moderate drops. These results suggest that the model relies on superficial lexical cues rather than robust semantic reasoning.

Dataset Type	Validation Accuracy (%)	Test Accuracy (%)
Contrast-Antonym	40	36
Contrast-Synonym	75	76
Contrast-Negation	66	64

Table 5: Accuracy drops on contrast validation and test sets across edit types.

After fine-tuning the model on the augmented training set containing both contrast examples and a 15% subset of the original SNLI training data, the model demonstrates marked recovery. The overall accuracy on the contrast validation set improves to **87%**, and on the contrast test set rises to **82%**. Detailed results by edit type are presented in Table 7.

SNLI Split	Baseline Model Accuracy (%)	Fine tune Model Accuracy with contrast set (%)
Validation set	77	87
Test set	76	82

Table 6: Comparison of finetuned and baseline model accuracy on contrast sets.

To test generalization, the fine-tuned model is also evaluated on the contrast test split derived from the SNLI test data split which is not used for model finetuning. It achieved better accuracy compared to the baseline model, confirming improved robustness to unseen perturbations .

Dataset Type	Validation Accuracy (%)	Test Accuracy (%)
Contrast–Antonym	62	54
Contrast–Synonym	84	76
Contrast–Negation	91	87

Table 7: Accuracy improvements after fine-tuning with contrast examples.

However, its accuracy remain somewhat lower than baseline model on test split shown in Table 6, indicating that synthetic edits may not always yield linguistically coherent examples to helps model learn the inherent linguistic relationship between premises and hypothesis. Hence, to assess the linguistic quality of the synthetic contrast hypotheses, I evaluated each edited hypothesis using a CoLA-based grammatical acceptability classifier (Warstadt et al., 2019; Morris et al., 2020)(model: *textattack/roberta-base-CoLA*). Table 8 summarizes the distribution of grammatical correctness by edit type.

Edit Type	% Correct	% Incorrect
Lexical Antonym	55.0%	45.0%
Lexical Synonym	51.3%	48.7%
Negation	95.7%	4.3%
None (Unedited)	96.8%	3.2%

Table 8: Grammatical statistics for the contrast validation hypotheses.

6 Analysis and Discussion

My analysis focuses on understanding how constrast set with different lexical and logical edits affect the robustness and uncertainty characteristics of the ELECTRA-small model trained on SNLI dataset. This section includes plots and figures related to the validation set, while the corresponding plots and figures for the test set are provided in Appendix B.

6.1 Baseline Model analysis

6.1.1 Accuracy Degradation Across Edit Types

The contrast dataset revealed distinct sensitivity patterns across perturbation categories. As shown in Table 5 and illustrated in Figure 1, *lexical antonym* edits induced the steepest accuracy drop (from 89% to 40%), followed by *negation* (66%) and *synonym* edits (75%). These results indicate that ELECTRA's token-level discrimination is

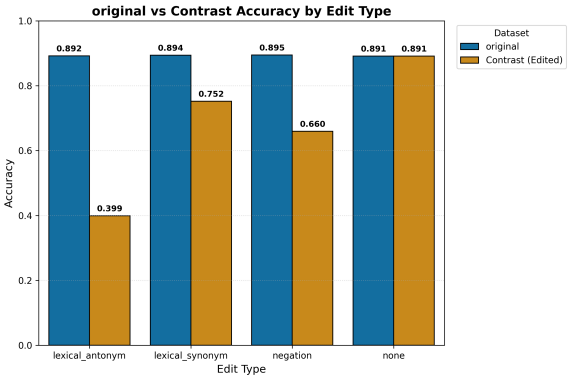


Figure 1: Baseline accuracy comparison across edit types (antonym, synonym, negation, and none/no-edits) for the original and contrast validation sets.

particularly brittle to polarity reversals and semantically subtle antonymic substitutions. This observation aligns with findings from Gardner et al. (2020), who showed that NLP models frequently rely on shallow lexical heuristics rather than genuine entailment reasoning.

Appendix B section B.1 illustrates accuracy drops in the SNLI test split.

6.1.2 Cartography-based Dataset Analysis

To better characterize model behavior, I employed the *data cartography* framework of Swayamdipta et al. (2020), which visualizes examples by their prediction confidence and variability. I extended this framework using *Monte Carlo Dropout as a Bayesian approximation* (Gal and Ghahramani, 2016) during evaluation (rather than training) to estimate uncertainty on the validation data. This adaptation expedite the model behavior data map analysis for pre-trained models, where full training-time uncertainty computation is computationally expensive and time consuming.

These uncertainty estimates are used to construct a cartography map plotting confidence (x-axis) versus variability (y-axis), color-coded by prediction correctness. Figure 2 compares data maps for original and contrast subsets. In the *unedited / none* subset remain similar in concentration and spread in both original and contrast set. The *negation*, *antonym* , and *synonym* subsets exhibit a pronounced shift toward the high-variability and ambiguous region, reflecting greater disagreement among stochastic forward passes which is inline with finding in Swayamdipta et al. (2020) cartography.



Figure 2: Baseline model datamap comparison across edit types (none/no-edits, negation, antonym, and synonym) for the original and contrast validation sets.

6.2 Effect of Fine-tuning with Contrast Data.

Fine-tuning with the augmented dataset (about 92k samples combining all edited examples and 15% of the original SNLI training set) mitigated the performance degradation. As detailed in Table 7 and illustrated in Figure 3, the model recovered to 87% accuracy on the contrast validation set, improving across all edit categories—most notably on negation (+25%). This demonstrates that explicitly exposing the model to contrastive examples improves robustness without substantial overfitting to the synthetic data.

6.3 Generalization to Unseen Data

When evaluated on a contrast-enhanced SNLI test split, the fine-tuned model achieved 82% accuracy versus 76% for the baseline, confirming improved robustness to unseen perturbations. However, the accuracy does not come close to its original, unedited test set (88%), suggesting that some synthetic edits may not be fully linguistically coherent. This trade-off between robustness and linguistic fidelity indicates that contrast data generation quality critically affects model performance.

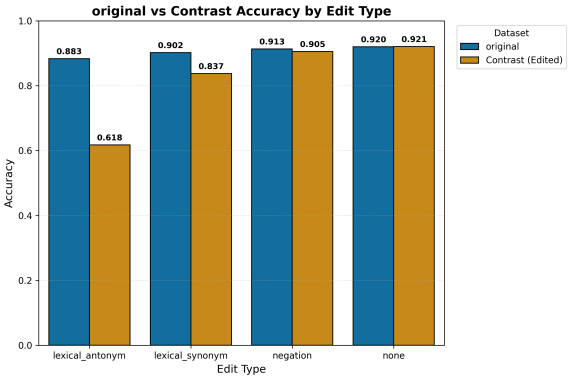


Figure 3: Fine-tuned model accuracy comparison across edit types (antonym, synonym, negation, and none/no-edits) for the original and contrast validation sets.

6.4 Grammatical Evaluation of the Contrast Set

The grammatical analysis indicates substantial variation in linguistic quality across edit types (Table 8). Lexical antonym and lexical synonym edits show relatively low acceptability (55.0% and 51.3%, respectively), suggesting that many automatic word substitutions introduced morphological or syntactic errors (e.g., agreement, incorrect inflection, or unnatural collocations). In contrast, negation edits exhibit high acceptability (95.7%), consistent with the observation that inserting a negation marker tends to be less disruptive to surface syntax than arbitrary lexical replacement.

The large share of flagged examples in the antonym and synonym categories plausibly explains part of why fine-tuning on the synthetic contrast set yielded only limited robustness gains. For instance, the lexical synonym "A woman tolerate in a line " was flagged as incorrect by the *CoLA-based classifier*, which is grammatical or morphological incorrect. Hence, next step is to explore instruction tuned LLM to generate more coherent perturbations.

6.5 Future Directions.

Future work will explore generating more coherent perturbations using instruction-tuned large language models such as *Mistral-7B-Instruct* to construct semantically valid contrast pairs. Techniques from Ross et al. (2022), Wu et al. (2021), and Dong et al. (2021) offer promising directions for linguistically grounded augmentation. Such improvements could better preserve entailment semantics while enhancing model invariance to lexical and logical variation.

7 Conclusion

In this work, I conducted a systematic robustness analysis of the *ELECTRA-small* model and then fine-tuned on the synthetically generated contrast set from *SNLI* dataset, focusing on model sensitivity to minimal semantic perturbations. By constructing a *contrast validation set* of about 10K examples—edited for negation, lexical antonyms, and synonyms, I observed substantial performance degradation relative to the unedited validation set. The model's accuracy dropped to 40% for antonym edits, 75% for synonym replacements, and 66% for negation, revealing its reliance on shallow lexical cues rather than true semantic inference.

Beyond accuracy analysis, I proposed a *modified data cartography framework* that applies *Monte Carlo dropout* during the validation phase, as a Bayesian approximation, to estimate prediction variability and model confidence. This adaptation enables training dynamics like insights without requiring access to full training trajectories, making it practical for models trained on large-scale data. The resulting data maps highlighted how contrast examples increase variability and ambiguity regions.

To address these weaknesses, I fine-tuned the model on a balanced dataset combining all contrast examples with portion of the original SNLI training data. This fine-tuning improved contrast validation set accuracy from 77% to 87%.

The findings demonstrate that *contrastive fine-tuning* can substantially enhance robustness without sacrificing overall accuracy. However, challenges persist for antonym and synonym edits, suggesting the need for more linguistically coherent premises-hypothesis pairs. Finally, the grammatical acceptability analysis using a *CoLA-based classifier* highlights limitations of rule-based synthetic edits and motivation for future exploration of instruction-tuned LLMs (e.g., *mistralai/Mistral-7B-Instruct-v0.1*) to generate higher-quality, semantically controlled contrastive examples.

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Appendix

A Implementation Commands

A.1 Baseline training

A.1.1 Training

```
python run.py --do_train --task nli --
    dataset snli --output_dir ./
    trained_model/
```

A.1.2 Evaluation

```
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_original_validation_9842_20251107065003
    .jsonl --model ./trained_model/model
    -baseline/ --output_dir ./
    eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_contrast_validation_9842_20251107065003
    .jsonl --model ./trained_model/model
    -baseline/ --output_dir ./
    eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_original_test_9824_20251107065322
    .jsonl --model ./trained_model/model
    -baseline/ --output_dir ./
    eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_contrast_test_9824_20251107065322
    .jsonl --model ./trained_model/model
    -baseline/ --output_dir ./
    eval_output/
```

A.2 Contrast set construction

```
python ./generate_contrast_set.py
    validation
python ./generate_contrast_set.py test
```

A.3 Data Analysis and evaluation

```
python ./dataset_analysis.py ./
    trained_model/model-baseline/ ./
    synthesized_data/
    snli_original_validation_9842_202511
    \07065003.jsonl ./synthesized_data/
    snli_contrast_validation_9842_202511
    \07065003.jsonl
```

```
python ./dataset_analysis.py ./
    trained_model/model-baseline/ ./
    synthesized_data/
    snli_original_test_9824_2025110706532
    \2.jsonl ./synthesized_data/
    snli_contrast_test_9824_20251107065322
    .jsonl
python ./dataset_analysis.py ./
    trained_model/model-finetune/ ./
    synthesized_data/
    snli_original_validation_9842_202511
    \07065003.jsonl ./synthesized_data/
    snli_contrast_validation_9842_202511
    \07065003.jsonl
python ./dataset_analysis.py ./
    trained_model/model-finetune/ ./
    synthesized_data/
    snli_original_test_9824_20251107065322
    .jsonl ./synthesized_data/
    snli_contrast_test_9824_20251107065322
    .jsonl
```

A.4 Performance Enhancement

A.4.1 fine tuning

```
python run.py --do_train --task nli --
    dataset snli --fine_tune yes --
    contrast_dataset ./synthesized_data/
    snli_contrast_
validation_9842_20251107065003.jsonl --
    model ./trained_model --output_dir
    ./trained_model/ --max_steps=12500
```

A.4.2 Evaluation

```
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_original_
validation_9842_20251107065003.jsonl --
    model ./trained_model/model-finetune
    / --output_dir ./eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_contrast_validation_9842_
    \20251107065003.jsonl --model ./
    trained_model/model-finetune/ --
    output_dir ./eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_original_test_9824\
    _20251107065322.jsonl --model ./
    trained_model/model-finetune/ --
    output_dir ./eval_output/
python run.py --do_eval --task nli --
    dataset ./synthesized_data/
    snli_contrast_test_9824\
    _20251107065322.jsonl --model ./
    trained_model/model-finetune/ --
    output_dir ./eval_output/
```

A.5 Analysis of synthetic contrast set

```
python analyze_grammar_contrast_set.py
```

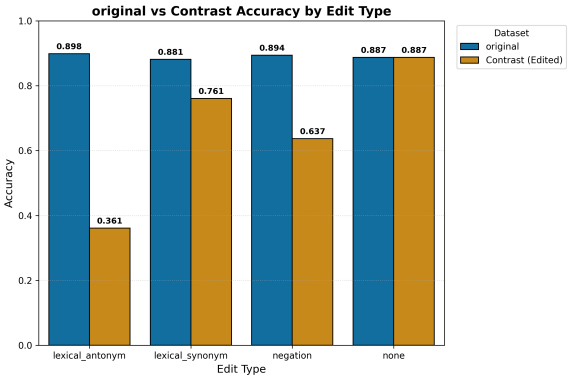


Figure 4: Baseline model accuracy comparison across edit types (antonym, synonym, negation, and none/no-edits) for the original and contrast test sets.

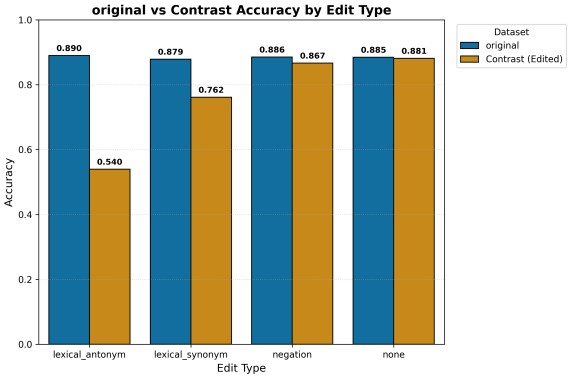


Figure 5: Finetune model accuracy comparison across edit types (antonym, synonym, negation, and none/no-edits) for the original and contrast test sets.

B Analysis and discussion

B.1 Baseline model accuracy comparison on test set

Figure 4

B.2 Finetune model accuracy comparison on test set

Figure 5